Check your progress

7.1 The nucleus of every cell contains several long threads, called **chromosomes**.

In most cells, there are **two** copies of each thread.

Each thread contains many **genes**.

The threads are made of a chemical called **DNA**.

7.2

Description	Egg cells	Sperm cells
contain a nucleus	✓	✓
are gametes	✓	✓
can swim		✓
in humans, contain 23 chromosomes	✓	✓
contain either an X chromosome or a Y chromosome		✓

7.3 **a** Sum = 112; 112 ÷ 20 = 5.6

b

Number of beans in a pod	3	4	5	6	7	8
Tally	𝍸	𝍸 |	𝍸 𝍸	𝍸 𝍸 |	𝍸 ||	𝍸 |||
Number of pods	2	3	5	4	3	3

c suitable scale on *x*-axis; suitable scale on *y*-axis; all bars correctly drawn.

7.4 **a** A substance that kills bacteria (but does not harm humans).

b The percentage of cases of penicillin-resistant pneumonia has increased.

c Any two from:

- This happened because penicillin was used to treat infections.
- Any bacteria that happened to be resistant to penicillin survived and reproduced.
- Their offspring inherited the genes that made them resistant.

If penicillin continues to be used over time, the proportion of bacteria that are resistant to penicillin increases.